Greedy Algorithms

1.

Consider the following scenario

You are designing a data compression system for a critical database server. You have analyzed the following character frequencies from the database logs:

Character	Frequency
A	50
B	15
C	20
D	10
E	5
F	5
G	30

Now answer the following questions:

a. **CO1** **Construct** a Huffman tree for the given characters and their frequencies. Additionally, calculate the space saved compared to using fixed 3-bit encoding for each character. Show your work step by step. **04**

b. **CO1** **Write** the Huffman Code for each character. **02**

c. **CO2** **Explain** how greedy property is used in Huffman coding and how it provides an optimal code. **02**

d. **CO2** Two friends are tasked with building the given Huffman tree. While both friends use priority queues in the process, **02**

- friend 1 inserts the newly created nodes at the end of the same frequency nodes.
- friend 2 inserts the newly created node at a random position between the same frequency nodes.

Explain which technique will provide more optimal code and why. If they are bound to provide equally optimal code, **explain** why.

2.

Consider the following scenario

You are designing a data compression system for a critical database server. You have analyzed the following character frequencies from the database logs:

Character	Frequency
A	45
B	13
C	12
D	16
E	9
F	9
G	12

Now answer the following questions:

a. **CO1** **Construct** a Huffman tree for the given characters and their frequencies. Additionally, calculate the space saved compared to using fixed 3-bit encoding for each character. Show your work step by step. **04**

b. **CO2** **Write** the Huffman Code for each character. **02**

c. **CO2** **Explain** how greedy property is used in Huffman coding and how it provides an optimal code. **02**

d. **CO2** Two friends are tasked with building the given Huffman tree. While both friends use priority queues in the process, **02**

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- friend 2 inserts the newly created node at a random position between the same frequency nodes.

Explain which technique will provide more optimal code and why. If they are bound to provide equally optimal code, **explain** why.

3.

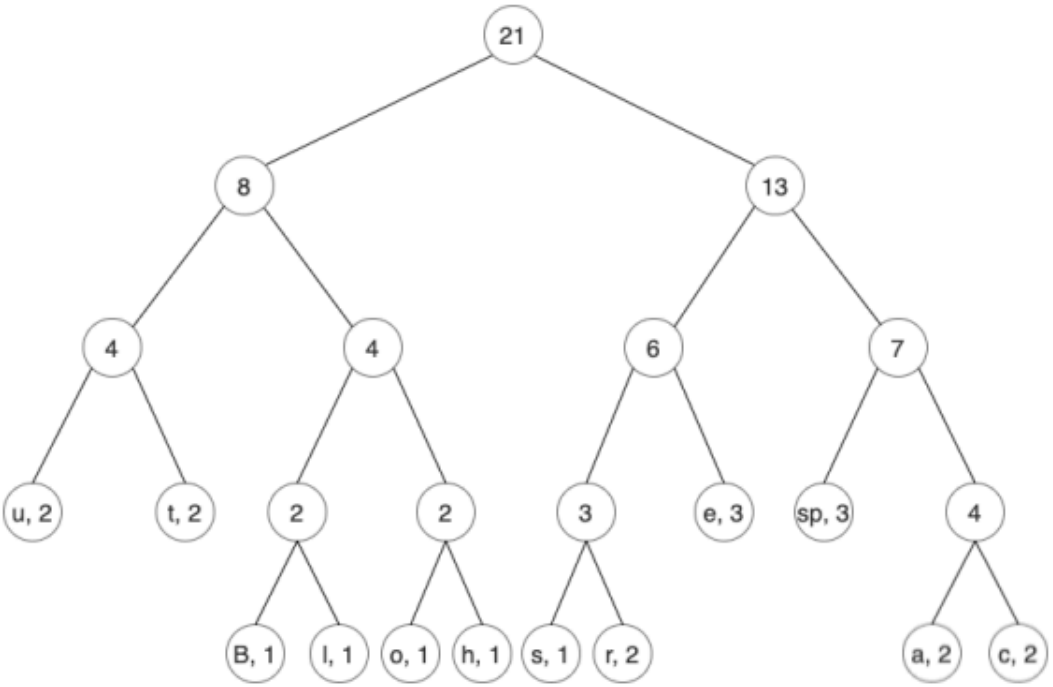
3		You are given two strings, '**cbabdcb**' and **'badcbc'**, and tasked with finding a common sequence with the longest length. Initially, you considered a recursive solution, but person 'X' suggested using a dynamic programming approach instead.	
	CO3	a. Find the longest common sequence using the approach proposed by person 'X'. **Show** the necessary **simulation** to find the answer.	**05**
	CO3	b. Additionally, you want to send these strings as an encoded message. After merging them into one string, **'cbabdcbbadcbc'**, now you need an efficient text-encoding algorithm with minimal space requirement. i. Simulate the encoding process, ii. construct any needed tables or trees, iii. generate the codeword for each character, iv. show the encoded message and v. the total number of bits required to store it.	**05**

4.

3		You are given two strings, '**abcdefg**' and **'bfegba'**, and tasked with finding a common sequence with the longest length. Initially, you considered a recursive solution, but person 'X' suggested using a dynamic programming approach instead.	
	CO3	a. Find the longest common sequence using the approach proposed by person 'X'. **Show** the necessary **simulation** to find the answer.	**05**
	CO3	b. Additionally, you want to send these strings as an encoded message. After merging them into one string, **'abcdefgbfegba'**, now you need an efficient text-encoding algorithm with minimal space requirement. i. Simulate the encoding process, ii. construct any needed tables or trees, iii. generate the codeword for each character, iv. show the encoded message and v. the total number of bits required to store it.	**05**

5.

b) [CO1] You are given the following tree generated by simulating the Huffman encoding algorithm on a text message.



i) **Write** the encoding for each of the characters. Consider left to be 0, and right to be 1 when traversing the tree. **04**

ii) Decode the following encoded message using the character codes you wrote on question (i). **01**

0100100111101111000010111101100010110110001011110111010011011000111100010 1

6.

2. Consider the following modified ASCII scheme and the encoded message using it.

Character	a	b	c	d	e	f	g	h	i	j
m_ascii	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001

Message encoded with m_ascii code:
0000 0100 0100 1000 0100 1000 0100 1000 0001 0101 0001 0001 0110 0000 0100 1001 0010 1001

Now answer the following questions.

a) [CO1] **Decode** the message. **01**

b) [CO1] **Show** simulation to find the encoding of each of the characters using Huffman coding technique. While creating a tree, less frequency nodes go to the left side of the root; left is considered as a 0, right is 1. **06**

c) [CO1] **Calculate** the number of bits required in the encoded message using huffman technique. **03**

7.

2. Consider the following modified ASCII scheme and the encoded message using it.

Character	a	b	c	d	e	f	g	h	i	j
m_ascii	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001

Message encoded with m_ascii code:
0010 0000 0010 0000 0011 0110 0111 1000 0011 1001 0000 0010 0011 0010 1001 1000 0010 0111

Now answer the following questions.

a) [CO1] **Decode** the message. **01**

b) [CO1] **Show** simulation to find the encoding of each of the characters using Huffman coding technique. While creating a tree, less frequency nodes go to the left side of the root; left is considered as a 0, right is 1. **06**

c) [CO1] **Calculate** the number of bits required in the encoded message using huffman technique. **03**

8.

2 a. CO2 You are given the following table containing symbols and their frequencies:

Symbol	A	B	C	D	+
Frequency	40	10	20	15	15

i) **Build** the Huffman code tree and find the codewords for each character. **04**
ii) **Decode** 100010111001010 using the Huffman code that you generated. **01**

b. CO4 You are given the arrival and departure times of eight trains for a railway platform, each in the following format: $[arrival\ time,\ departure\ time)$. Only one train can use the platform at a time.

Suppose, you have got the following train-use requests for the next day.

$$\{\,[8,\ 13),\ [6,\ 9),\ [11,\ 14),\ [2,\ 7),\ [1,\ 7),\ [12,\ 20),\ [7,\ 13)\,,\ [13,\ 20)\,\}$$

i) **Find** the maximum number of trains that can use the platform without any collision. **03**
ii) **Determine** the minimum number of platforms that needs to be there to ensure the arrival and departure of all these trains without collision. **02**

9.

2 a. CO2 You are given the following table containing symbols and their frequencies:

Symbol	A	B	C	D	#
Frequency	35	15	25	10	20

i) **Build** the Huffman code tree and find the codewords for each character. **04**
ii) **Decode** 100010111001010 using the Huffman code that you generated. **01**

b. CO4 You are given the arrival and departure times of eight trains for a railway platform, each in the following format: $[arrival\ time,\ departure\ time)$. Only one train can use the platform at a time.

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ii) **Determine** the minimum number of platforms that needs to be there to ensure the arrival and departure of all these trains without collision. **02**

10.

Alice and Bob both are the peers of a decentralized distributed network. Alice wants to send a message to Bob using the platform but she does not want that any other users of the network would be able to understand the message. That is why she encodes the message using an encoding algorithm which is known to only both of them and sends it to Bob. The message is:

Abaaccdd efkkslasae,eEkls aaad.

a. CO4	**Simulate** the Huffman Encoding algorithm to encode the message. Construct Huffman Tree, generate the codeword for each character, show the encoded message and count the total number of bits required to store the message.	**7**
b. CO5	**Explain** the advantage of Huffman Encoding over the ascii codes approach mathematically.	**1**
c. CO4	After reading the message Bob replied to Alice by sending an encoded message: 100010101011111101010. Now **use** the Huffman Tree constructed in question 2(a) to decode the message.	**2**

11.

Alice and Bob both are the peers of a decentralized distributed network. Alice wants to send a message to Bob using the platform but she does not want that any other users of the network would be able to understand the message. That is why she encodes the message using an encoding algorithm which is known to only both of them and sends it to Bob. The message is:

Saaees eklassddd,Aeeesss lkes.

a. CO4	**Simulate** the Huffman Encoding algorithm to encode the message. Construct Huffman Tree, generate the codeword for each character, show the encoded message and count the total number of bits required to store the message.
b. CO5	**Explain** the advantage of Huffman Encoding over the ascii codes approach mathematically.
c. CO4	After reading the message Bob replied to Alice by sending an encoded message: 1111110000111011010. Now **use** the Huffman Tree constructed in question 2(a) to decode the message.

12.

4		You are now attending the CSE221 final exam and there is a limited time to solve the questions. Each question requires a different amount of time and offers varying marks. After answering some questions, you suddenly realized that only 6 minutes remained. You aim to maximize the total marks you can score, given the constraints on time. Questions and their properties: ▪ Question 4: Time = 4 minutes, Marks = 7 ▪ Question 5: Time = 2 minutes, Marks = 9 ▪ Question 6: Time = 3 minutes, Marks = 10 ▪ Question 7: Time = 5 minutes, Marks = 8 ▪ Question 8: Time = 6 minutes, Marks = 12 ▪ Question 9: Time = 1 minute, Marks = 3	
a. **CO3**		If your faculty doesn't give partial marks for incomplete answers, which combination of questions maximizes your total marks without exceeding the time limit? Hurry up, you need a quick, efficient approach—no time for brute force. **Show** a full detailed calculation.	**04**
b. **CO3**		Fortunately, your faculty enters the hall and announces a new rule: partial marks will be given for partially completed answers. If you can solve part of a question, you'll get marks proportional to the time you spend on it. For example, solving half of a 4-minute question gives you half the marks. Now, with this new opportunity, how does your strategy shift? How will you maximize your score within the time limit, factoring in that you can now earn something even if you don't fully complete a question? **Calculate** the maximum marks you can score with this fortunate announcement. Did you score more than the previous scenario?	**04**
c. **CO2**		How does Dynamic Programming have a lower time complexity than brute force or recursive solutions? Explain briefly.	**02**

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Questions and their properties:

- Question 4: Time = 5 minutes, Marks = 8
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- Question 7: Time = 6 minutes, Marks = 10
- Question 8: Time = 2 minutes, Marks = 6
- Question 9: Time = 1 minute, Marks = 5

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c. **CO2**	How does Dynamic Programming have a lower time complexity than brute force or recursive solutions? Explain briefly.	**02**

14.

Wye is a CSE BracU Student. The pre-advising week is approaching, and he is trying to find the most suitable course combination for next semester. As per his previous academic results, his credit limit **has been set to 9**. Different courses have different credits. He came up with values for these courses by consulting with students who have already taken them. He wants to select the most valued courses. Courses, credits and values are provided below:

Course	PHY112	CSE330	CSE331	CSE370	CSE499
Credit	1	2	3	4	5
Value	9	7	15	12	16

CO1 a. Wye wants to try all possible combinations to find the best one. Help him by simulating a suitable dynamic programming approach. **Show** your work in detail, then **write** down the selected courses.

CO1 b. Zed, a friend of Wye, adopted a slightly different approach and considered the courses based on their per-credit value. **Determine** whether Zed could select more valued courses than Wye.
Note that, selecting a course still means taking the whole, with full credits.

CO1 c. If Zed is successful, he will send a message **Successful**; otherwise, **Unsuccessful**. Instead of fixed-length coding, he is willing to send the message using Huffman Coding. **Show** simulation of Huffman coding to encode his message.

15.

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16.

Following are the codes generated from a text for a Huffman tree construction.

H - 1000 *o* - 1001 <*space*> - 1010 *S* - 1011	*u* - 000 *d* - 001 *n* - 010	*e* - 011 *l* - 110 *t* - 111

You are also given the following information:

- The frequency of each leaf node except *e*, *l*, and *t* is 1.
- The left and right child nodes of the root have frequencies 5 and 8 respectively.

Now answer the following questions.

CO2

a) Suppose in a Huffman tree, the distances from the root to the pair of leaves denoting the letters *k* and *b* are 5 and 2 respectively, which letter between them is more frequent in the original text? Just **mention** the letter. **01**

b) **Draw** the Huffman tree from the given coding table above. **03**

c) Continuing on Q(b), what are the frequencies of *l*, and *t* in the original text? Just **mention** the frequencies. **02**

Your lazy friend, Tom, used a very simple encoding method for compressing a text file. The text file contains the string : *Sudden Hello*. He used a constant number of bits for encoding each of the ***n*** distinct characters in this text. The constant is $ceil(log_2 n)$ in this case. On the other hand, you have compressed the text file using your own huffman tree from Q(b).

d) **Mention** how many bits each of the characters in Tom's encoding scheme contain. **01**

e) **Compare** between the number of bits needed to decode the above-mentioned string using your friend's scheme and your Huffman tree from Q(b). **03**

17.

Following are the codes generated from a text for a Huffman tree construction.

J - 1000 *o* - 1001 <*space*> - 1010 *W* - 1011	*n* - 000 *h* - 001 *m* - 010	*l* - 011 *e* - 110 *c* - 111

You are also given the following information:

- The frequency of each leaf node except *e*, *l*, and *c* is 1.
- The left and right child nodes of the root have frequencies 5 and 8 respectively.

Now answer the following questions.

CO2

a) Suppose in a Huffman tree, the distances from the root to the pair of leaves denoting the letters *m* and *e* are 3 and 5 respectively, which letter between them is more frequent in the original text? Just **mention** the letter. **01**

b) **Draw** the Huffman tree from the given coding table above. **03**

c) Continuing on Q(b), what are the frequencies of *e*, and *c* in the original text? Just **mention** the frequencies. **02**

Your lazy friend, Tom, used a very simple encoding method for compressing a text file. The text file contains the string : *Welcome John*. He used a constant number of bits for encoding each of the ***n*** distinct characters in this text. The constant is $ceil(log_2 n)$ in this case. On the other hand, you have compressed the text file using your own huffman tree from Q(b).

d) **Mention** how many bits each of the characters in Tom's encoding scheme contain. **01**

e) **Compare** between the number of bits needed to decode the above-mentioned string using your friend's scheme and your Huffman tree from Q(b). **03**

18.

You Study in a university and there are some faculties in your university and their class schedule (start and end time) are given below. As a pantomath (who wants to learn a lot) person you want to gather more knowledge and do class of the maximum faculties possible. But, due to the time conflict, one student cannot cover all the classes. So, you asked your friends Derke, Leo and Chronicle how you can maximize the number of classes. Derke said you should choose the classes which have the lowest duration and Leo said you should choose those classes which end early and Chronicle said you should pick classes that start early.

a) [CO1] Who's method will you **select** in order to complete the classes of maximum faculties? **01**

b) [CO1] Using the method you mentioned in (a) **simulate** and find out the maximum number of classes you can do from the following schedule and also find out the initials of the faculties. **06**

Faculty	Start Time	End Time
MIBA	1	10
MZU	2	3
AGD	6	8
MNR	10	11
RIM	5	7
FGZ	3	6
SBD	7	10

c) [CO1] As one student cannot cover all the classes so you want to determine the minimum number of students needed so that all the classes are covered. **Explain** your algorithm in a pseudocode/ flow-chart/ step-by-step instructions format. **03**

19.

You Study in a university and there are some faculties in your university and their class schedule (start and end time) are given below. As a pantomath (who wants to learn a lot) person you want to gather more knowledge and do class of the maximum faculties possible. But, due to the time conflict, one student cannot cover all the classes. So, you asked your friends Derke, Leo and Chronicle how you can maximize the number of classes. Derke said you should choose the classes which have the lowest duration and Leo said you should choose those classes which end early and Chronicle said you should pick classes that start early.

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b) [CO1] Using the method you mentioned in (a) **simulate** and find out the maximum number of classes you can do from the following schedule and also find out the initials of the faculties. **06**

Faculty	Start Time	End Time
MIBA	2	11
MZU	3	4
MNR	7	9
AGD	11	12
FGZ	6	8
MNR	4	7
SBD	8	11

c) [CO1] As one student cannot cover all the classes so you want to determine the minimum number of students needed so that all the classes are covered. **Explain** your algorithm in a pseudocode/ flow-chart/ step-by-step instructions format. **03**

20.

a) [CO1] Welcome to the exciting realm of Bracula Gaming World, where you can earn lots of money by playing different games. But there's a twist – time is limited to just 20 hours! Each game has its own time limit and prize money if played for the full duration. However, you can play for a **fraction of the time** and still earn money in proportion. Check the list of games below.
Your task is to make the most of your 20 hours and collect **as much money as possible**.

Game Name	**Total Time Limit**	**Prize Money For Full Duration**
Road Rush	15 hours	75$
GTA Vice City	20 hours	90$
Virtual Cop	3 hours	30$
House of the Dead	6 hours	55$
Far Cry	2 hours	17$
Devil May Cry	10 hours	47$

i) **Name** the algorithm tailored for this situation. **01**
ii) **Write** the names of the games and their respective playing times that will lead to maximum prize money. Show a simulation of your algorithm. **04**

b) [CO1] You are given the following tree generated by simulating the Huffman encoding algorithm on a text message.

21.

a) [CO1] Welcome to the exciting realm of Bracula Gaming World, where you can earn lots of money by playing different games. But there's a twist – time is limited to just 20 hours! Each game has its own time limit and prize money if played for the full duration. However, you can play for a **fraction of the time** and still earn money in proportion. Check the list of games below.
Your task is to make the most of your 20 hours and collect **as much money as possible**.

Game Name	**Total Time Limit**	**Prize Money For Full Duration**
Road Rush	16 hours	85$
GTA Vice City	20 hours	95$
Virtual Cop	4 hours	30$
House of the Dead	7 hours	45$
Far Cry	3 hours	18$
Devil May Cry	11 hours	50$

i) **Name** the algorithm tailored for this situation. **01**

ii) **Write** the names of the games and their respective playing times that will lead to maximum prize money. Show a simulation of your algorithm. **04**

22.

The final exam has just ended, and you are on a well-deserved trip to a remote area. During your journey, you have collected different types of digital content that you want to share with your friends and family. Each *Content Type* has a *Total Duration* and a *Value*. Total Duration measures the content's pre-compression size, while Value reflects its importance or meaning to the audience.

Due to bandwidth constraints, one of your friends suggested using Huffman encoding, a lossless compression technique that assigns shorter binary codes to more voluminous (i.e., longer duration) items.

Content Type	Total Duration (Seconds)	Value
A	14	21
B	12	15
C	10	7
D	9	18
E	7	12
F	6	8
G	5	7

a. **CO1** Construct a Huffman Tree based on the given **Total Duration** to assign optimal binary codes to each content type. **04**

b. **CO1** Calculate the Total Number of Bits *(duration * code length)* required to transmit the full content of each type. **03**

c. **CO1** Again, due to bandwidth constraints, you are now trying to select which content to send and how much of each. You can shorten any content, but its value drops proportionally if you do so. Your goal is to choose the types and lengths of the contents that give you the highest total value without going over *35 seconds*. **Determine** which content to send and how long each part should be. **03**